

# SUSHANT LOKHANDE

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## EXPERIENCE

**San Jose State University** | Graduate Research Assistant, Systems Testing & Validation Aug 2025 to May 2026

- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Automated provisioning, configuration, and **telemetry** for containerized test clusters, cutting experiment setup effort **55%** and making every sweep reproducible from config.
- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.

**Turito** | Software Engineering Intern, Reliability & Performance Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Lowered network egress **14%** by removing redundant media-path processing and simplifying routing and payload handling.

**Trivia Softwares** | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.
- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

## TECHNICAL SKILLS

**Languages:** Python, C++17, Rust, Java, SQL, Bash

**Test & Validation:** test automation frameworks, pytest, Locust load testing, regression suites, interoperability and failure-recovery scenarios, failure classification, root-cause analysis

**Systems & Tooling:** Linux, Docker, Git, CI/CD, raw-socket networking, CPU and lock-contention profiling, telemetry pipelines, structured logging

**Data & Analysis:** NumPy, Pandas, SQLite, statistical significance testing, dashboards, reproducible experiment configs

## PROJECTS

**Proxima, Vector Search Engine** | C++17, AVX2/SIMD, HNSW, pybind11

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.

**TCP Sliding Window Simulator** | Python, raw sockets, Linux

- Implemented handshake, sliding window, cumulative ACKs, and retransmission over raw sockets with live packet-state visualization.

## OPEN SOURCE

- fission/fission:** 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration
- NVIDIA/garak:** preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)

## EDUCATION

**San Jose State University** | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

*Coursework:* Computer Networks, Distributed Computing, Advanced Algorithms, Database Systems, Machine Learning, Artificial Intelligence

**University of Mumbai** | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

*Research:* malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)